

Dendrite-Free Lithium Anode Enables the Lithium//Graphite Dual-Ion Battery with Much Improved Cyclic Stability

Xiao-Tong Xi,[†] Wen-Hao Li,[†] Bao-Hua Hou,[†] Yang Yang,[†] Zhen-Yi Gu,[†] and Xing-Long Wu^{*,†,‡,§}

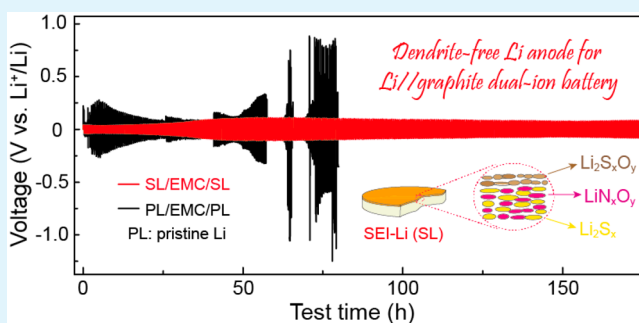
[†]National & Local United Engineering Laboratory for Power Batteries, Faculty of Chemistry, Northeast Normal University, Changchun, Jilin 130024, People's Republic of China

[‡]Key Laboratory for UV Light-Emitting Materials and Technology, Northeast Normal University, Changchun, Jilin 130024, People's Republic of China

Supporting Information

ABSTRACT: Owing to the high de/intercalation potential of anions on graphite cathode, dual-ion battery (DIB) has attracted much attention for advanced energy storage. Among various DIBs, Li//graphite (Li//G) configuration delivers the highest voltage to achieve the theoretically highest energy density due to the lowest Li-plating/-stripping potential. Unfortunately, the dendrite nature of Li anode brings a series of serious problems, fatally restricting the practical application of Li//G DIB. Herein, we for the first time propose fabricating artificial and robust a solid electrolyte interphase layer on Li anode to suppress Li dendritic growth during cycling and hence to improve the cyclic stability of Li//G DIB. As a result, a long-life Li//G DIB is assembled and achieved successfully with the enhanced and smoothed Coulombic efficiencies (CE). For example, the Li//G DIB exhibits a high capacity retention of 89.6% with an average CE of 97.8% and initial discharge capacity of 95.2 mAh g⁻¹ over 500 cycles at 1 A g⁻¹. This study points out a new possible strategy to enhance the energy-storage performance of Li//G DIB with high practicability.

KEYWORDS: dual-ion battery, lithium anode, dendrite-free, solid electrolyte interphase, long cycle life



Lithium-ion batteries based on the electrochemical reactions of single-cation (Li⁺) de/intercalation have been extensively used to power the mobile society for decades, although their energy densities are still too low to satisfy the increasing demands of various consumer electronics and power tools.¹ Hence, new-type batteries are persistently being sought to ensure the progressiveness and endurance of the battery.^{2–4} Recently, dual-ion battery (DIB) has attracted great attention due to the high anion de/intercalation potential on cathode^{5–7} that is one of the key prerequisites to achieve high energy densities. For example, the commonly used graphite cathode can de/intercalate PF₆⁻ at a high voltage of up to 5.2 V vs Li⁺/Li due to its stable layer structure with the relatively large interlayer space of 0.34 nm.⁸ However, there are still a lot of challenges for DIB before practical application. In addition to the relatively low specific capacity of anion storage in the cathode materials presently, another critical problem is the very poor cyclic stability.

Constant efforts have been made for the anode materials of DIB, such as soft carbon,⁹ mesocarbon microbeads,¹⁰ Al,^{11,12} Sn,^{13,14} and so on. In comparison to these insertion-/alloying-type anode materials, lithium-anode-based Li//graphite (Li//G) DIB can realize the highest working voltage owing to the lowest redox potential of the Li-plating/stripping reaction. Unfortunately, the extremely high activity of metallic

lithium usually brings a series of serious side effects between Li anode and electrolyte, fatally lowering the Coulombic efficiency (CE) and cycle life of Li//G DIB. In addition, the dendritic growth of Li anode as well as huge volume changes during the successive cycling would be likely to prick through the separator, leading to high safety risks (such as short circuit and even explosion) and hence restricting the practicability of Li-anode-based batteries including Li//G DIB.

Recently researchers have explored the failure mechanisms of metallic Li anode^{15,16} and proposed some effective strategies to deal with them in other Li-anode-based batteries, although these strategies have not been used for Li//G DIB. The strategies to stabilize the Li anode usually include the following: using appropriate electrolytes and additives,^{17,18} designing anode structure,^{19,20} exploiting solid electrolyte,^{21,22} and building a stable Li/electrolyte interface.^{23,24} In general, the solid electrolyte interphase (SEI) films spontaneously generated between Li anode and electrolyte are structurally fragile and compositionally heterogeneous. Hence, artificial SEI with robust stability and high ionic conductivity is highly expected to passivate and enhance the surface of Li anode via

Received: October 15, 2018

Accepted: December 7, 2018

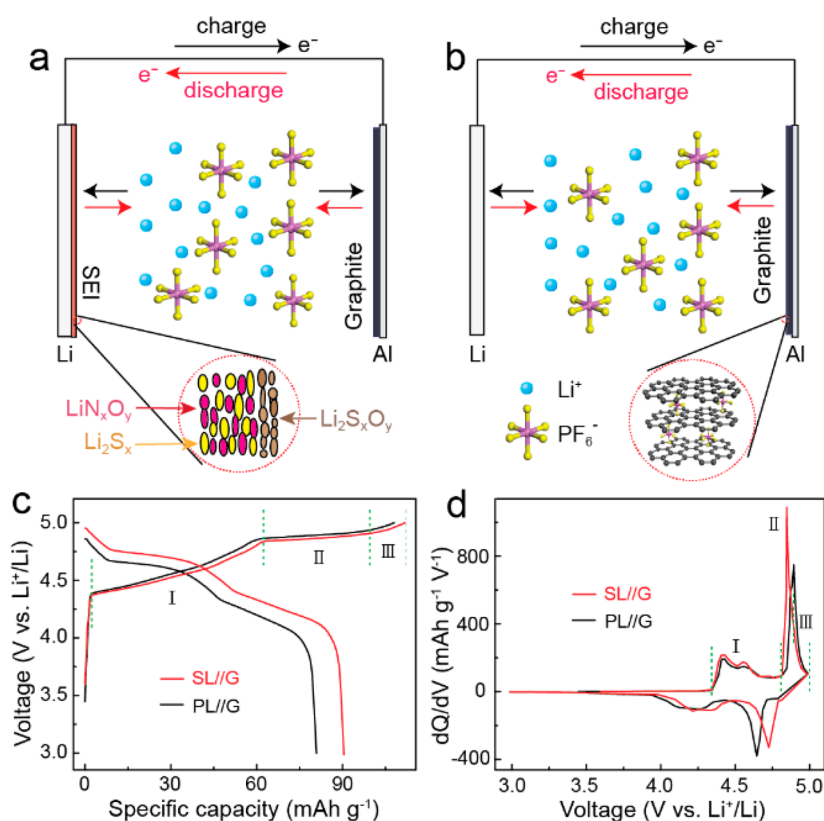


Figure 1. Schematic illustrations of the (a) SL//G and (b) PL//G DIB. Inset in panel a shows the heterogeneous constitution of the designed *h*-SEI film on Li anode. (c) Galvanostatic charge–discharge curves at 0.2 A g^{−1} and (d) corresponding dQ/dV differential curves of the SL//G and PL//G DIB, respectively.

optimizing the composition of electrolyte and utilizing the efficient additives in the electrolyte. Previously, it has been demonstrated that lithium nitrate (LiNO₃) additive in electrolyte can effectively minimize the shuttle effect of polysulfides and hence improve the cyclic stability of Li–S battery although such LiNO₃-assisted SEI cannot suppress the Li dendritic growth.²⁵ In further study for Li–S battery, Cui and co-workers have confirmed that a dendrite-repressible SEI can be fabricated on Li anode under the cooperative action of the deliberately added LiNO₃ and lithium polysulfide in electrolyte.²⁶ Nevertheless, there is still less study of utilizing such Li-dendrite-suppressing strategy to improve the energy-storage performance of Li-anode-based DIB. Herein, we rationally propose for the first time that constructing robust and conductive SEI film with the Li-dendrite-repressible capability on Li anode would be also highly effective in enhancing the electrochemical properties of Li//G DIB, because it is one type of typical metallic Li anode based battery. As a demonstration, the Li//G DIB based on artificial SEI-modified Li anode exhibits the Li-dendrite-free nature during cycling and hence much enhanced cyclic stability in terms of higher capacity retention (e.g., 98.3% vs 77.6% over 150 cycles at 1 A g^{−1}, 1 C corresponding to the current density of 0.1 A g^{−1}) as well as smoother and higher Coulombic efficiencies, compared to the pristine Li-anode-based one.

Details for coating the Li-dendrite-repressible SEI on Li foils are presented in the experimental section in the [Supporting Information](#), which is similar to those in previous reports.²⁶ Briefly, the symmetric Li/electrolyte/Li cells are cycled in a ternary electrolyte that is LiTFSI (lithium bis-(trifluoromethanesulfonyl)imide, 1 mol L^{−1})–LiNO₃ (0.5

mol L^{−1})–Li₂S₈ (0.18 mol L^{−1}) in 1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME) at the volume ratio of 1:1 ([Figure S1](#)). As studied in detail by Cui and co-workers,²⁶ the reduction products (LiN_xO_y and Li₂S_x) of LiNO₃ and Li₂S₈ are simultaneously deposited on the Li surface at the beginning of cycling. Afterward, the polysulfides in the electrolyte will be further oxidized to form the more dense lithium sulfates (Li₂S_xO_y) by LiNO₃ and deposited above the former layer, generating the heterogeneous SEI (*h*-SEI; inset in [Figure 1a](#)) coated on the Li anode. In addition, there are also organic composites of ROCO₂Li and ROLi generated from the decomposition of solvents through the whole *h*-SEI.²⁷ It is found that the surfacial dense degree of the formed SEI is highly related to the key formation parameters, viz., current density (*j*) and the time of one single charge/discharge (CD) process. As the scanning electron microscope (SEM) images show in [Figure S2](#), too short CD time (0.25 h, [Figure S2a](#)) leads to the formation of uneven and porous SEI, and too long one (1 h, [Figure S2c](#)) will roughen the SEI with easy breakability. In addition, SEI growth is too fast at a high *j* of 2.0 mA cm^{−2} ([Figure S2f](#)), resulting in lots of loose holes in the SEI layer on the Li surface, while the SEI film generated at a too low *j* of 0.1 mA cm^{−2} ([Figure S2d](#)) is also rugged due to the excessive side reactions. As a result, optimal *j* and single CD time to construct a dense, homogeneous, and well-distributed SEI coating on Li electrode ([Figure S2e](#) and [Figure S3](#)) should be 0.2 mA cm^{−2} and 0.5 h, respectively. Additionally, [Figure S4](#) compares the surfacial morphologies of pristine Li (PL, the corresponding elemental mappings are shown in [Figure S5](#)) electrode with the optimal SEI-modified

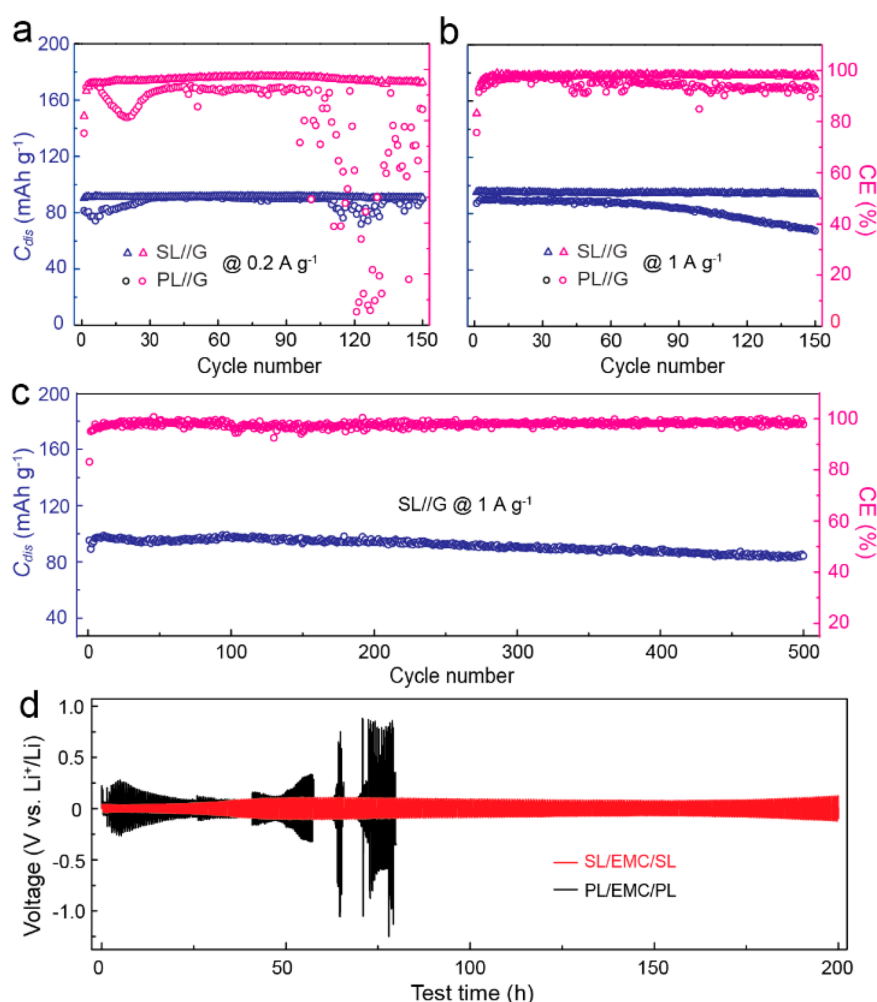


Figure 2. Comparison of cycling performance between SL//G and PL//G at various current densities of (a) 0.2 and (b) 1 A g^{-1} . (c) Long-life cycling performance of SL//G DIB over 500 cycles at 1 A g^{-1} . (d) Voltage profiles vs time in the Li symmetric cells. The electrolyte is 4 mol L^{-1} LiPF_6 in EMC. Black and red indicate PL/EMC/PL and SL/EMC/SL cells, respectively.

Li (SL) one, further confirming the unique coating of artificial *h*-SEI layer on Li foil because of the increased coarseness.

In order to confirm the capability of SL anode to enhance the energy-storage performance of Li//G DIB, two DIB configurations (schematically illustrated in Figure 1a,b), SL//graphite (SL//G) and PL//graphite (PL//G), are assembled and compared first by galvanostatic CD (GCD) in the electrolyte of 4 mol L^{-1} LiPF_6 in ethyl methyl carbonate (EMC). As shown in Figure 1a,b, both exhibit a similar working mechanism. Exactly, during the charging processes, the anions of PF_6^- in electrolyte intercalate into graphite cathode, and Li^+ cations in the electrolyte are electrochemically reduced to metallic state which then deposits on the Li counter electrode. Conversely, both PF_6^- anions and Li^+ cations transfer back into the electrolyte in the discharging process due to the deintercalation and stripping reactions, respectively. The only difference between them is the presence of the aforementioned artificial *h*-SEI on the Li anode in SL//G DIB, which can suppress the Li dendritic growth during the successive Li-plating/stripping processes. Figure 1c compares the first GCD curves of SL//G and PL//G in a voltage window of 3–5 V. Both exhibit the typical and similar three-step CD profiles due to the PF_6^- de/intercalation from/into the interlayers of graphite.¹¹ Taking the charge curve of SL//G

as an example, the three steps are marked as I, II, and III, according to the voltage variations from low to high. Each region corresponds to the stepwise PF_6^- intercalation in the graphite cathode, which has been studied previously in detail.²⁸ The dQ/dV differential curves (Figure 1d) are further used to discriminate the two CD processes. The main difference is that, SL//G delivers much lower polarization (the voltage difference between cathodic and anodic peaks) compared to PL//G, implying its enhanced electrochemical kinetics in SL//G due to the improved Li surface (artificial *h*-SEI on SL vs the spontaneously formed SEI during the initial cycles on Li in PL//G²³). In addition to the better CD patterns, SL//G delivers the higher discharge capacity (C_{dis}) of 90.5 mAh g^{-1} than 81.2 mAh g^{-1} for PL//G at the close charge capacity, suggesting the increased cyclic reversibility in SL//G DIB. In detail, SL//G has a smaller voltage hysteresis than PL//G, which is owing to the compact and uniform *h*-SEI film on SL that stabilizes the Li/electrolyte interfaces and inhibits the decomposition of electrolyte, resulting in faster charge transfer (Figure 3a). Nevertheless, the SEI film chemically/electrochemically in situ formed in EMC on PL is loose and rugged; the unstable SEI layer will lead to the continuous decomposition of electrolyte and become thicker and thicker with cycles, further increasing the voltage polarization for PL//

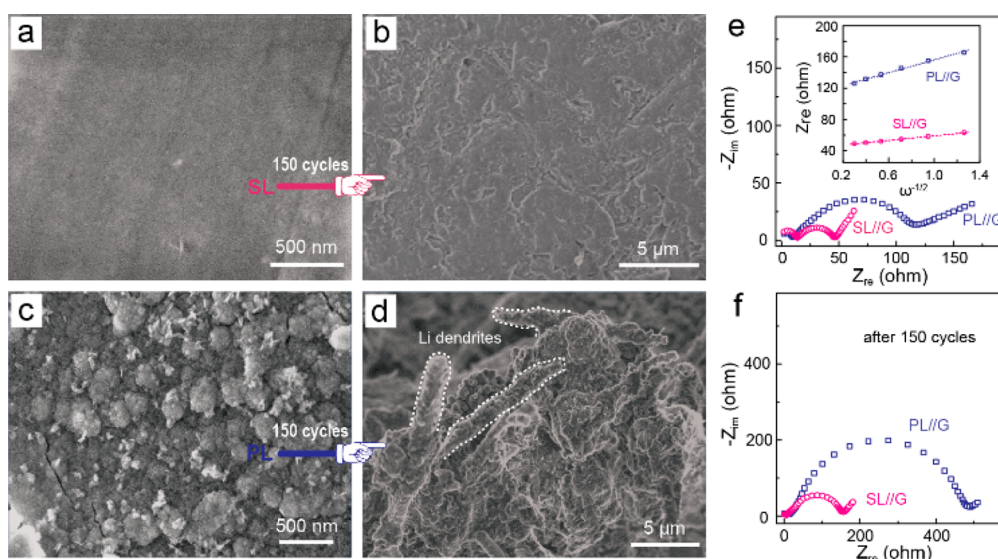


Figure 3. Surface SEI images of (a, b) SL and (c, d) PL anodes (a, c) before cycling and (b, d) after 150 cycles in the corresponding Li//G DIB. Note that electrode c is the one immersed in the electrolyte of 4 mol L⁻¹ LiPF₆/EMC over 12 h to stabilize its surface and to ensure the in situ formation of chemical SEI. (e) Nyquist plots of impedance spectra of SL//G and PL//G before cycling. Inset shows the linear fitting between Z_{re} and reciprocal square root of frequency ($\omega^{-1/2}$) in the low-frequency region for SL//G and PL//G, respectively. (f) Impedance spectra of SL//G and PL//G after 150 cycles. All EIS plots are collected at the open circuit voltage of 4.8 V vs Li⁺/Li with an amplitude voltage of 5 mV.

G (Figure 3c). Note that the SL anode used in the present SL//G is also obtained at 0.2 mA cm⁻² with a single CD time of 0.5 h. This parameter is the optimized one as compared in Figures S6 and S7 (cycling performance and electrochemical impedance, respectively), which is consistent with the aforementioned observations of SEM images.

Panels a and b of Figure 2a further compare the cycling performance of SL//G and PL//G. As shown, SL//G DIB always exhibits the much better cyclic stability, in terms of higher capacity retention with the larger C_{dis} and higher CE values with smoother patterns, in comparison to the PL//G one. For example, the PL//G DIB becomes dead only after ~90 cycles at 0.2 A g⁻¹ (Figure 2a) which is judged from the chaotic C_{dis} and CE variations above 90 cycles, while the SL//G delivers a much more stable cycling characteristic. At a larger j of 1 A g⁻¹ over 150 cycles (Figure 2b), SL//G exhibits a capacity retention of 98.3%, which is obviously higher than 77.6% for PL//G. More significantly, SL//G always delivers the higher CE values than PL//G in the whole cycling processes including the first cycle (83.4% vs 77.2%) and the following cycles (e.g., 97.5–99% vs 90–93.5% at 130–150 cycles), implying that the side reactions in Li//G DIB have been effectively suppressed due to the artificial *h*-SEI coating on Li anode. In addition, SL//G also exhibits long-life cyclic stability. As shown in Figure 2c, the capacity retention is up to 89.6% over 500 cycles (corresponding to a very low capacity decay of 0.021% per cycle) with an average CE of 97.8% and initial C_{dis} of 95.2 mAh g⁻¹ at 1 A g⁻¹. From the comparison between SL//G and PL//G, it can be concluded that there is no obvious structure change in graphite structure, and the capacity decay caused by graphite can be almost ignored during 150 cycles.

The above-disclosed enhancement of SL//G DIB for energy storage strongly indicates that the artificial *h*-SEI film has indeed improved the interface characteristics of Li anode via suppressing Li dendritic propagation and retarding the continuous break/repair of SEI film on the Li anode surface, which has been verified below. First, Li symmetric cells

assembled from different Li electrodes were tested and compared at a constant j of 0.5 mA cm⁻² in 4 mol L⁻¹ LiPF₆/EMC electrolyte (Figure 2d). As shown, the PL/EMC/PL cell exhibits much larger potential fluctuation during cycling, and its polarization becomes extremely high with the anodic/cathodic voltage of 0.75–1 V abruptly after 70 h, indicating the rapid and massive formation of Li dendrites on the Li electrode. In comparison, the SL/EMC/SL cell gives the much more stable polarized potentials of ±120 mV after even 200 h. Furthermore, ex situ SEM tests directly observe the dendrite-free nature of SL anode during cycling (Figure 3a–d). As shown, the SL surface is still very neat and smooth without the distinct Li dendritic growth (Figures 3b and S8) after 150 cycles at 0.2 A g⁻¹. In comparison, there are already lots of needle-like Li dendrites (1.5–2.5 μm in diameter and tens of micrometers in length, shown in Figures 3d and S9) on the PL anode cycled in PL//G DIB at the same condition. These suggest that the artificial *h*-SEI film can effectively prevent the generation of Li dendrites, while the loose and rugged SEI (Figures 3c and S10) chemically/electrochemically in situ formed in ester electrolyte on PL is almost incapable of suppressing Li dendritic growth. Additionally, the *h*-SEI on SL should be highly stable against the EMC solvent, guaranteeing the successful practice of metallic Li anode for Li//G DIB.

In addition to suppressing the growth of Li dendrites, the coated *h*-SEI film on Li can also improve the interfaces between Li anode and electrolyte and hence promote the kinetics of electrochemical reactions, which is demonstrated by the following studies of electrochemical impedance spectroscopy (EIS) and cyclic voltammetry (CV). Figure 3e first shows the EIS patterns of the freshly assembled SL//G and PL//G. Apparently, both consist of two semicircles at the high and medium frequency, belonging to the resistances of SEI layer (R_{SEI}) and charge transfer (R_{ct}), respectively, and a straight line at low frequency that is assigned to the Warburg impedance (Z_w) of ion diffusion. The R_{ct} (~32 Ω) in SL//G is much smaller than that (~105 Ω) in PL//G, although SL//G owns slightly higher R_{SEI} compared to PL//G. This indicates that *h*-

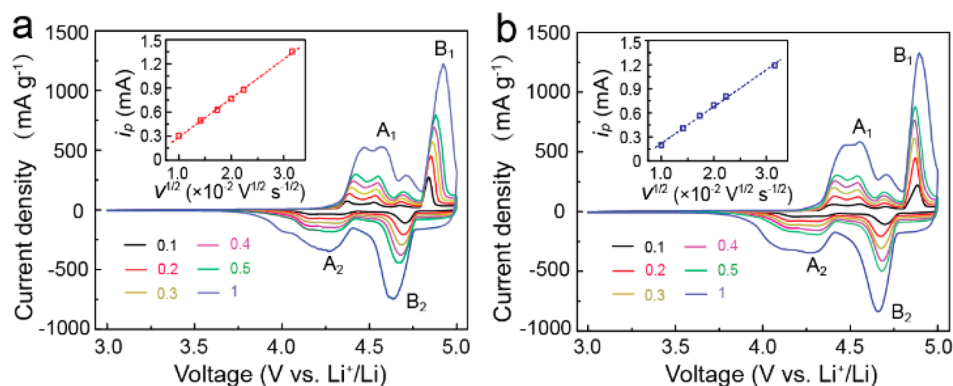


Figure 4. CV curves of (a) SL//G and (b) PL//G at various scan rates from 0.1 to 1.0 mV s^{-1} in the voltage range of 3–5 V. Two insets show the linear fitting between i_p and $v^{1/2}$ for peak B_1 to calculate the D_{app} values.

SEI coated on Li anode can effectively facilitate the charge (ions and electrons) transfer processes at the interfaces without the obvious increase of film resistance. After 150 cycles, the total resistance ($R_{\text{SEI}} + R_{\text{ct}}$) of PL//G has increased to $\sim 490 \Omega$, which is also obviously higher than $\sim 150 \Omega$ for SL//G (Figure 3f), implying that h -SEI can prevent the continuous decomposition of electrolyte and stabilize the Li anode during cycling. In essence, this reveals that the change of impedance spectra is closely related to the SEI surface morphology on the Li surface for Li//G DIB (Figure 3a–d).

Studies on the anionic diffusion are also meaningful because the anion insertion/extraction into/from the interlayers of graphite is one of the key electrochemical processes related in DIB and determines its energy-storage performance. Hence, the apparent diffusion coefficients (D_{app}) are further calculated and discussed via fitting the EIS plots. The inset in Figure 3e shows the linear fitting results between Z_{re} and reciprocal square root of frequency ($\omega^{-1/2}$) in the low-frequency region. D_{app} values can be evaluated by the following eqs 1 and 2:

$$D = (R^2 T^2) / (2 A^2 n^4 F^4 C^2 \sigma^2) \quad (1)$$

$$Z_{\text{re}} \propto \sigma \omega^{-1/2} \quad (2)$$

The detailed calculation processes are shown in the Supporting Information. According to the calculation results, the D_{app} value in SL//G is $5.17 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$, which is ~ 7.4 times larger than $0.70 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$ in PL//G. A value of D_{app} so much larger for SL//G compared to PL//G implies that the optimized surface of Li anode by artificial h -SEI can synergistically promote the electrochemical kinetics of PF_6^- insertion/extraction on the graphite cathode and the whole electrochemical reactions.

Figure 4 shows the CV curves of SL//G and PL//G at various scan rates of 0.1, 0.2, 0.3, 0.4, 0.5, and 1 mV s^{-1} in the voltage window of 3–5 V vs Li^+/Li . According to previous reports,²⁹ the peak shape and peak current (i_p) evolution in CV curves at various scan rates is related to the kinetics of ion insertion/extraction, and the relationship between i_p and scan rate (v) can be fitted by the following eq 3:

$$\log(i_p) = b \log(v) + \log(a) \quad (3)$$

where a and b are two constants. From the fitting results, it is found that two B_1 peaks (corresponding to PF_6^- insertion into graphite) in both cells are mainly the diffusion-controlled process, because both b values are close to 0.5.²⁹ As the insets show in Figure 4a,b, all i_p values of peak B_1 in two cells exhibit

a good linear dependence on the respective square root of scan rates ($v^{1/2}$), also indicating their diffusion-controlled behavior. Furthermore, the D_{app} values in SL//G and PL//G were calculated according to the Randles–Sevcik equation,³⁰ which is also shown in the Supporting Information in detail. Coinciding with the fitting results of EIS plots, the D_{app} value in SL//G is also larger than that in PL//G ($9.68 \times 10^{-10} \text{ cm}^2 \text{ s}^{-1}$ vs $9.10 \times 10^{-10} \text{ cm}^2 \text{ s}^{-1}$), although the difference between them is smaller than the former one obtained from EIS fitting. Moreover, as can be seen from Figure 4a,b, the peaks in the CV patterns have no obvious deviation, indicating that artificial h -SEI film is thin and conductive. Hence, the h -SEI has little impact on the voltage platform for SL//G DIB.

The great improvement of cyclic stability including the higher capacity retention as well as higher and smoother CE in the Li//G DIB should be at least attributed to the following four factors. First, the artificial h -SEI film generated at the joint action of Li_2S_8 and LiNO_3 additives exhibits the high mechanical strength²⁶ to prevent the dendrite growth on the Li anode surface and to maintain a stable interphase during cycling. Second, the generated h -SEI covers the entire surface of Li anode, greatly minimizing the continuous decomposition of electrolyte during the Li-plating/stripping processes. Third, such h -SEI film also improves the interfaces between Li anode and electrolyte, which can effectively decrease the electrochemical impedance of the whole Li//G DIB. Fourthly, the optimization of Li anode surface by h -SEI coating has synergistically promoted the kinetics of the electrochemical reactions, which has been demonstrated by the increased D_{app} values calculated from both EIS and CV.

In summary, we have proposed a new and effective strategy of Li anode modification by artificial SEI film to highly enhance the cyclic stability of Li//G DIB in common carbonate electrolyte. As a result, the developed DIB delivers a higher C_{dis} (95.2 mAh g^{-1} at 1 A g^{-1}) and working volage with the much improved cycling performance (a capacity retention of up to 89.6% after 500 cycles at 1 A g^{-1}) as well as higher and smoother CE values, compared to the pristine Li-based one. It has also been verified by the studies of ex situ SEM and electrochemical kinetics that the dendrite-free nature of modified Li anode and the improved interfaces between Li and electrolyte play crucial roles in enhancing the energy-storage performance. In view of the high effectiveness in terms of performance enhancement, this study points out a new possible strategy to promote the practicability of advanced Li//G DIB.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsam.8b01764.

Experimental details; calculations of apparent diffusion coefficient; SEM and element mappings analysis (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: xinglong@nenu.edu.cn.

ORCID

Xing-Long Wu: 0000-0003-1069-9145

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the National Natural Science Foundation of China (Grant No. 51602048) and the Fundamental Research Funds for the Central Universities (Grant No. 2412017FZ013).

■ REFERENCES

- (1) Goodenough, J. B.; Park, K. S. The Li-ion rechargeable battery: a perspective. *J. Am. Chem. Soc.* **2013**, *135*, 1167–1176.
- (2) Lu, J.; Li, L.; Park, J. B.; Sun, Y. K.; Wu, F.; Amine, K. Aprotic and aqueous Li-O₂ batteries. *Chem. Rev.* **2014**, *114*, 5611–5640.
- (3) Guo, J.-Z.; Wu, X.-L.; Wan, F.; Wang, J.; Zhang, X.-H.; Wang, R.-S. A Superior Na₃V₂(PO₄)₃-Based Nanocomposite Enhanced by Both N-Doped Coating Carbon and Graphene as the Cathode for Sodium-Ion Batteries. *Chem. - Eur. J.* **2015**, *21*, 17371–17378.
- (4) Guo, J.-Z.; Yang, Y.; Liu, D.-S.; Wu, X.-L.; Hou, B.-H.; Pang, W.-L.; Huang, K.-C.; Zhang, J.-P.; Su, Z.-M. A Practicable Li/Na-Ion Hybrid Full Battery Assembled by a High-Voltage Cathode and Commercial Graphite Anode: Superior Energy Storage Performance and Working Mechanism. *Adv. Energy Mater.* **2018**, *8*, 1702504.
- (5) Rodríguez-Pérez, I. A.; Jian, Z.; Waldenmaier, P. K.; Palmisano, J. W.; Chandrasekhar, R. S.; Wang, X.; Lerner, M. M.; Carter, R. G.; Ji, X. A Hydrocarbon Cathode for Dual-Ion Batteries. *ACS Energy Lett.* **2016**, *1*, 719–723.
- (6) Wang, M.; Tang, Y. A Review on the Features and Progress of Dual-Ion Batteries. *Adv. Energy Mater.* **2018**, *8*, 1703320.
- (7) Li, W.-H.; Ning, Q.-L.; Xi, X.-T.; Hou, B.-H.; Guo, J.-Z.; Yang, Y.; Wu, X.-L.; Chen, B. Highly Improved Cycling Stability of Anion De-/Intercalation in the Graphite Cathode for Dual-Ion Batteries. *Adv. Mater.* **2018**, DOI: 10.1002/adma.201804766.
- (8) Read, J. A.; Cresce, A. V.; Ervin, M. H.; Xu, K. Dual-graphite chemistry enabled by a high voltage electrolyte. *Energy Environ. Sci.* **2014**, *7*, 617–620.
- (9) Fan, L.; Liu, Q.; Chen, S.; Xu, Z.; Lu, B. Soft Carbon as Anode for High-Performance Sodium-Based Dual Ion Full Battery. *Adv. Energy Mater.* **2017**, *7*, 1602778.
- (10) Ji, B.; Zhang, F.; Wu, N.; Tang, Y. A Dual-Carbon Battery Based on Potassium-Ion Electrolyte. *Adv. Energy Mater.* **2017**, *7*, 1700920.
- (11) Zhang, X.; Tang, Y.; Zhang, F.; Lee, C.-S. A Novel Aluminum-Graphite Dual-Ion Battery. *Adv. Energy Mater.* **2016**, *6*, 1502588.
- (12) Tong, X.; Zhang, F.; Chen, G.; Liu, X.; Gu, L.; Tang, Y. Core-Shell Aluminum@Carbon Nanospheres for Dual-Ion Batteries with Excellent Cycling Performance under High Rates. *Adv. Energy Mater.* **2018**, *8*, 1701967.
- (13) Sheng, M.; Zhang, F.; Ji, B.; Tong, X.; Tang, Y. A Novel Tin-Graphite Dual-Ion Battery Based on Sodium-Ion Electrolyte with High Energy Density. *Adv. Energy Mater.* **2017**, *7*, 1601963.
- (14) Wang, M.; Jiang, C.; Zhang, S.; Song, X.; Tang, Y.; Cheng, H. M. Reversible calcium alloying enables a practical room-temperature rechargeable calcium-ion battery with a high discharge voltage. *Nat. Chem.* **2018**, *10*, 667–672.
- (15) Cheng, X. B.; Zhang, R.; Zhao, C. Z.; Zhang, Q. Toward Safe Lithium Metal Anode in Rechargeable Batteries: A Review. *Chem. Rev.* **2017**, *117*, 10403–10473.
- (16) Guo, Y.; Li, H.; Zhai, T. Reviving Lithium-Metal Anodes for Next-Generation High-Energy Batteries. *Adv. Mater.* **2017**, *29*, 1700007.
- (17) Zeng, X.-X.; Yin, Y.-X.; Shi, Y.; Zhang, X.-D.; Yao, H.-R.; Wen, R.; Wu, X.-W.; Guo, Y.-G. Lithiation-Derived Repellent toward Lithium Anode Safeguard in Quasi-solid Batteries. *Chem.* **2018**, *4*, 298–307.
- (18) Zhang, X. Q.; Chen, X.; Cheng, X. B.; Li, B. Q.; Shen, X.; Yan, C.; Huang, J. Q.; Zhang, Q. Highly Stable Lithium Metal Batteries Enabled by Regulating the Solvation of Lithium Ions in Nonaqueous Electrolytes. *Angew. Chem., Int. Ed.* **2018**, *57*, 5301–5305.
- (19) Yan, K.; Lu, Z.; Lee, H.-W.; Xiong, F.; Hsu, P.-C.; Li, Y.; Zhao, J.; Chu, S.; Cui, Y. Selective deposition and stable encapsulation of lithium through heterogeneous seeded growth. *Nat. Energy* **2016**, *1*, 16010.
- (20) Yang, C. P.; Yin, Y. X.; Zhang, S. F.; Li, N. W.; Guo, Y. G. Accommodating lithium into 3D current collectors with a submicron skeleton towards long-life lithium metal anodes. *Nat. Commun.* **2015**, *6*, 8058.
- (21) Yang, L.; Wang, Z.; Feng, Y.; Tan, R.; Zuo, Y.; Gao, R.; Zhao, Y.; Han, L.; Wang, Z.; Pan, F. Flexible Composite Solid Electrolyte Facilitating Highly Stable “Soft Contacting” Li-Electrolyte Interface for Solid State Lithium-Ion Batteries. *Adv. Energy Mater.* **2017**, *7*, 1701437.
- (22) Zhang, W.; Nie, J.; Li, F.; Wang, Z. L.; Sun, C. A durable and safe solid-state lithium battery with a hybrid electrolyte membrane. *Nano Energy* **2018**, *45*, 413–419.
- (23) Li, N. W.; Yin, Y. X.; Yang, C. P.; Guo, Y. G. An Artificial Solid Electrolyte Interphase Layer for Stable Lithium Metal Anodes. *Adv. Mater.* **2016**, *28*, 1853–1858.
- (24) Xu, R.; Zhang, X.-Q.; Cheng, X.-B.; Peng, H.-J.; Zhao, C.-Z.; Yan, C.; Huang, J.-Q. Artificial Soft-Rigid Protective Layer for Dendrite-Free Lithium Metal Anode. *Adv. Funct. Mater.* **2018**, *28*, 1705838.
- (25) Liang, X.; Wen, Z.; Liu, Y.; Wu, M.; Jin, J.; Zhang, H.; Wu, X. Improved cycling performances of lithium sulfur batteries with LiNO₃-modified electrolyte. *J. Power Sources* **2011**, *196*, 9839–9843.
- (26) Li, W.; Yao, H.; Yan, K.; Zheng, G.; Liang, Z.; Chiang, Y. M.; Cui, Y. The synergistic effect of lithium polysulfide and lithium nitrate to prevent lithium dendrite growth. *Nat. Commun.* **2015**, *6*, 7436.
- (27) Xiong, S.; Xie, K.; Diao, Y.; Hong, X. Characterization of the solid electrolyte interphase on lithium anode for preventing the shuttle mechanism in lithium–sulfur batteries. *J. Power Sources* **2014**, *246*, 840–845.
- (28) Noel, M.; Santhanam, R. Electrochemistry of graphite intercalation compounds. *J. Power Sources* **1998**, *72*, 53–65.
- (29) Zhang, K.; Park, M.; Zhou, L.; Lee, G.-H.; Li, W.; Kang, Y.-M.; Chen, J. Urchin-Like CoSe₂ as a High-Performance Anode Material for Sodium-Ion Batteries. *Adv. Funct. Mater.* **2016**, *26*, 6728–6735.
- (30) Tang, K.; Yu, X.; Sun, J.; Li, H.; Huang, X. Kinetic analysis on LiFePO₄ thin films by CV, GITT, and EIS. *Electrochim. Acta* **2011**, *56*, 4869–4875.